Group Project Milestone 1:
Literature Review and Exploratory Data Analysis

Predicting Delivery Performance and Customer Satisfaction

An analysis of 99,441 orders from the Olist Brazilian e-commerce marketplace

Group 10

Pan Pinyou	A0329105L
Li Yudan	A0330335R
Zhu Gengyu	A0333773Y

National University of Singapore

IT5006 Fundamentals of Data Analytics

AY2026/2027 Semester 1

1 Literature Review

1.1 Scope

This project analyses the *Olist Brazilian E-Commerce* dataset: nine relational tables covering 99,441 orders placed on a Brazilian marketplace between September 2016 and October 2018 [11]. Olist is a two-sided marketplace – it aggregates small sellers onto Brazil's large retail platforms and handles fulfilment – so each order links a customer, one or more sellers, a basket, a payment, a delivery and usually a review. That structure supports all three question families the brief identifies: operational performance, customer experience and commercial outcomes.

This review covers what Phase 2 will depend on: what drives fulfilment quality and satisfaction, which model families suit tabular transactional data, how to engineer features from a relational order model without leaking the target, and which metrics survive severe class imbalance.

1.2 Fulfilment quality and customer satisfaction

The e-commerce service-quality literature is unusually direct about which lever matters. Parasuraman et al. [12] decompose electronic service quality into efficiency, *fulfilment*, system availability and privacy, and find fulfilment – the accuracy of delivery promises and the condition in which goods arrive – among the strongest contributors to perceived quality. Rao et al. [14] sharpen this into logistics terms, showing across 260 online retailers that order-fulfilment quality feeds directly into purchase satisfaction and subsequent retention.

The implication for a transactional extract is that delivery is not merely an operational statistic; it is the dominant *observable* input to how a customer rates the transaction. Our analysis reproduces that link sharply (Section 2.5), and it is why our second candidate problem is framed after delivery rather than at checkout. The same literature also warns what is *not* observable here: perceived quality is mediated by product expectations, seller communication and returns handling, none of which appears in an order table. Any performance ceiling we report must be read against that omission.

1.3 Model families for tabular transactional data

For structured data of this size the practical model space is well characterised. Regularised linear and logistic models are the natural baseline: cheap, interpretable, and – as Section 2.6 shows – revealing about distribution shift. Tree ensembles are the usual step up. Breiman [2] reduce variance by bagging decorrelated trees, while gradient boosting [5] and its scalable implementations [4] fit residuals stagewise with shrinkage and explicit complexity penalties. Grinsztajn et al. [7] benchmark tree ensembles against modern deep architectures across 45 datasets and find tree-based methods remain state of the art at this scale, attributing the gap to inductive biases better matched to irregular tabular targets.

We therefore adopt a small model budget: one linear family (ridge and ℓ_2-regularised logistic regression) and one tree ensemble (histogram-based gradient boosting), reused across both framings. The brief asks that complexity be justified by evidence, and Section 2.6 supplies it in both directions – the ensemble earns its place on lead time, while on repeat purchase neither family beats the baseline meaningfully.

1.4 Feature engineering on a relational order model

Deriving features from nine tables raises a problem a flat file does not. Three of them – items, payments and reviews – are one-to-many against orders, so a naive join multiplies rows and reweights every aggregate towards multi-item orders. The remedy is to aggregate each child relation to the analysis grain first and only then join.

Beyond that the usual transactional vocabulary applies: counts and sums over the basket, ratios that normalise for order size, calendar encodings, and geographic distance from postcode centroids. High-cardinality categoricals such as the 74 product categories are poorly served by one-hot encoding; target encoding [10] compresses them while retaining signal, provided it is fitted inside the cross-validation loop.

Leakage is the dominant risk. Kaufman et al. [9] formalise it as information about the target that would not legitimately be available at prediction time, and identify the two forms that threaten this dataset: post-outcome features and improper partitioning. Critically, their framing makes leakage relative to a *prediction moment* – the principle our feature policy is built on, since delivery outcomes are forbidden when forecasting delivery but admissible when predicting a review not yet written. On partitioning, Bergmeir and Benítez [1] show that standard k-fold cross-validation is invalid for temporally ordered data.

1.5 Evaluation under imbalance and drift

Both classification targets here are heavily imbalanced, and the brief is explicit that accuracy alone misleads. Saito and Rehmsmeier [15] show that precision–recall curves are more informative than ROC curves under strong imbalance, because ROC-AUC is insensitive to a large true-negative mass. We report PR-AUC against its no-skill floor alongside balanced accuracy, treating raw accuracy as a diagnostic rather than a headline. Class weighting and resampling such as SMOTE [3] are standard remedies, though both shift the effective prior.

Regression brings a second complication. Hyndman and Koehler [8] show that many accuracy measures degenerate in common cases and argue for judging errors against a naive benchmark. That is decisive here, because delivery performance improves dramatically across the window (Section 2.3). Under such a level shift, R^2 against a baseline fitted on the past turns negative even for a useful model, whereas mean absolute error stays interpretable. This is textbook concept drift [6], whose central recommendation – that models over non-stationary data be evaluated and periodically refitted with respect to time – shapes our Phase 2 plan. All modelling uses `scikit-learn` pipelines [13], keeping preprocessing inside each fold.

1.6 Conclusion: trends, gaps and opportunities

Three themes emerge. The service-quality literature converges on *fulfilment* as the dominant observable driver of e-commerce satisfaction. The tabular machine-learning literature converges on a small set of model families, with tree ensembles as the reasonable ceiling at this scale. And the methodological literature is emphatic that on imbalanced, temporally ordered data the choice of split and metric determines whether a number means anything.

The gap we address is narrow but concrete. Most work on this marketplace is descriptive, or predicts review scores using features that would not exist when the prediction is needed. Little of it is explicit about *when* a model would run – and that choice changes both what is admissible and what is achievable, by a factor of two in our own results. Making the prediction moment a first-class design decision, and measuring what it costs, is the contribution of Section 2.6.

2 Exploratory Data Analysis

2.1 Dataset overview and data model

The dataset comprises **nine** related CSV tables totalling roughly 1.45 million rows. `orders` is the spine, at 99,441 rows – one per order – and the other eight attach to it directly or through `order_items`. Figure 1 shows the recovered structure and Table 3 gives the full data dictionary.

Three tables are **one-to-many** against orders, and joining them naively is the first trap the data sets.[a]

The second trap is identity. `customer_id` is regenerated for every order, so it is an order key rather than a person: the customers table holds 99,441 distinct `customer_id` values but only **96,096** distinct `customer_unique_id` values. Grouping on the wrong one reports zero repeat buyers; grouping on the right one finds 2,997 people (3.1%) who ordered more than once, one of them 17 times. We carry `customer_unique_id` throughout and split on it (Section 2.6).

2.2 Data quality

Table 1 records every issue and our decision; the decisions live in `src/data_load.py`, so the code and this table cannot drift apart.

Missingness here is **structural rather than random**. 97.0% of orders reach status *delivered*; the remaining 3.0% are shipped, cancelled, unavailable or still processing, and 2.98% therefore carry no delivery timestamp. That is not a defect to impute – the event has not happened – so delivery targets are undefined for those orders and they are excluded from delivery models. Elsewhere the data is remarkably complete: 99.2% of orders in the analysis window carry a review, and product attributes are missing for under 2% of products.

Two further decisions shape the analysis window. The platform's first four months are a pilot contributing 326 orders in total, against a 2018 run rate above 6,000 per month, and the final weeks are truncated mid-cycle. We restrict the analysis to **2017-01-01 to 2018-08-31**, leaving 99,091 orders, and show the ramp-up separately (Figure 2).

2.3 Temporal patterns: growth, seasonality and drift

Monthly order volume roughly triples across the window. The single largest event is **Black Friday, November 2017**, at 7,288 orders, about 1.5 times a typical month (Figure 3a). Purchasing follows a clear weekly and daily rhythm: weekday-heavy, peaking at 16:00, with 22% of orders placed at weekends (Figure 4).

The most consequential temporal finding is not seasonality but **drift**. Mean delivery lead time falls from 16.9 days in February 2018 to 7.7 days in August 2018: Olist materially improved its fulfilment over twenty months (Figure 3b). Across the chronological split used later, the mean falls by **5.34 days** between the training and test periods.

Lateness behaves differently again – it is **volatile rather than trending**, swinging from 1.4% (June 2018) to 21.4% (March 2018). Those spikes align with Black Friday congestion and with the nationwide truckers' strike that paralysed Brazilian road freight in May 2018: exogenous shocks that no order-level feature available at checkout can anticipate.

Together these two facts set the evaluation strategy. Under a level shift of this size, R^2 measured against a baseline fitted on the past is negative even for a genuinely useful model, so we report **mean absolute error** as the headline regression metric [8], and we treat periodic refitting as a Phase 2 requirement rather than an optimisation [6].

2.4 Customers, sellers, products and geography

The marketplace is geographically **lopsided**, and that asymmetry is the engine of the delivery problem: **59.7%** of sellers sit in São Paulo state against **42.0%** of customers. Only 35.7% of

[a] **On tables and joins.** A direct orders–items–payments merge yields 118,437 rows rather than 99,441, a 19% inflation, and one order expands into as many as 63 rows. Every mean computed on that frame is silently reweighted towards large, multi-item, multi-payment orders. We aggregate each child table to one row per order first (counts, sums, and the attributes of the most expensive item) and only then join, asserting the grain afterwards.

orders are fulfilled within the customer's own state, and the median customer–seller distance is **434 km** (Figure 5). Distance is in turn the strongest checkout-time correlate of lead time ($r = 0.40$), ahead of the platform's own delivery estimate ($r = 0.38$).

Demand is long-tailed across 74 product categories, with the top ten covering roughly two-thirds of orders. The typical order is small – median item value R$86.90 – but **freight is not a rounding error**: the median order pays R$17.17 in freight, about 22% of item value. Payment behaviour is characteristically Brazilian, with 75.4% of orders paid by credit card and 51.4% split across more than one instalment (Figure 6).

2.5 Reviews, and the relationship that dominates

Review scores are severely imbalanced – **77.1%** are four or five stars and 57.8% are five – exactly the imbalance the brief warns about. The interesting structure is what produces the minority.

A late delivery raises the probability of a 1–2 star review from **9.2%** to **54.0%**, a **5.9-fold** increase, and the effect is monotone in how late the parcel is (Figure 7). This is the clearest relationship in the dataset and it reproduces the fulfilment–satisfaction link that Rao et al. [14] identify. It also has a direct methodological consequence: a model predicting review score *at checkout* cannot see the variable that matters most, because it has not happened yet.

2.6 Candidate problem exploration

We examined three candidates. All probes use `scikit-learn` pipelines on a chronological split (train to 2018-04-30, validation to 2018-06-30, test thereafter) that is additionally **grouped on `customer_unique_id`**, removing the 227 buyers who would otherwise appear on both sides.[b]

Leakage depends on when the model runs. Rather than one global rule we define a forbidden set per **prediction moment**. *At checkout*, nothing about fulfilment or the customer's opinion exists: delivery dates, review scores and even `order_approved_at` are all future. *At delivery*, the parcel has arrived and we are predicting a review not yet written, so the delivery outcome is history and is admissible – the brief itself lists "delivery performance" among the features for this problem. The guard `assert_no_leakage(features, regime)` enforces the correct set and every modelling cell must pass it.

The cost of getting this wrong is measurable. Adding one forbidden column, the realised delivery time, to the late-delivery model lifts ROC-AUC from **0.589 to 1.000** and balanced accuracy to 0.988 (Figure 9). That is not a result; it is the target restated.

Candidate 1 — delivery performance, framed twice (primary). One question answered as a regression on lead time in days and as a classification of late versus on-time, satisfying the brief's requirement for both task types from a single coherent problem. Stakeholder: operations.

The regression works. Gradient boosting reaches **MAE 3.89 days** against 6.65 for a constant fitted on the training period, a **41.5%** reduction, with ridge regression at 4.61 days in between. Notably, Olist's own delivery promise is a *worse* point forecast than the constant (MAE 9.79 days), because the platform pads its estimates deliberately – a promise the customer beats is good service. It remains a useful feature, just not a useful prediction.

The classification is weak, and Section 2.3 explains why. Balanced accuracy is 0.580 and PR-AUC 0.121 against a 0.074 floor, a 1.6× lift. Crucially, **always predicting "on time"**

[b] **Note on splitting.** Chronology alone is insufficient: 2,997 people placed more than one order, so a purely temporal cut can still put one of a person's orders in training and another in test. This is the concrete form of the brief's `customer_id` versus `customer_unique_id` warning, and it is why we also report a test subset containing only customers never seen in training.

scores 0.926 accuracy while catching not a single late order – the exact trap the brief describes.

Candidate 2 — low review score, predicted at delivery (secondary). The strongest classifier we found: PR-AUC **0.402** against a 0.110 floor (3.6×), balanced accuracy **0.692**, ROC-AUC 0.730. Performance on customers never seen in training is identical (PR-AUC 0.403), so the model is not memorising individual buyers.

The prediction moment is what makes it work. The same target predicted *at checkout*, without delivery information, reaches only PR-AUC 0.212: the delivery experience nearly doubles the achievable signal, quantifying the relationship in Section 2.5. Stakeholder: customer experience, with a concrete action – flag an at-risk order for outreach before the review lands.

Candidate 3 — repeat purchase within 30 days (rejected). This fails twice over. The base rate is only 1.8% – Olist is overwhelmingly a one-off marketplace, with 96.9% of customers appearing exactly once – and the probe reaches balanced accuracy 0.517, essentially chance. Separately, **right-censoring destroys the evaluation**: at the more natural 90-day horizon *no* labelled orders remain in the test window at all, since the final 90 days cannot have their outcome observed. That is a structural limit of a 21-month extract, not something a better model repairs. Table 2 scores all three candidates.

2.7 Key insights

1. Three of the nine tables are one-to-many against orders; joining naively inflates the data by 19% and biases every average towards large orders.
2. `customer_id` is an order key, not a person. The distinction is the difference between reporting 0% and 3.1% repeat buyers.
3. Delivery performance **drifts**: mean lead time halves over the window and falls 5.34 days between the training and test periods, which is why MAE rather than R^2 is the honest regression metric.
4. Lateness is **volatile, not predictable** – 1.4% to 21.4% monthly, driven by Black Friday and the May 2018 truckers' strike. This bounds the late-delivery classifier at roughly 1.6× the no-skill floor.
5. A late delivery raises the 1–2 star rate from 9.2% to 54.0%, a 5.9× increase and the strongest relationship in the data.
6. **The prediction moment is a design decision, not a detail.** Moving the review model from checkout to post-delivery nearly doubles PR-AUC (0.212 to 0.402); moving it the other way, by admitting one forbidden column, inflates ROC-AUC to a meaningless 1.000.

We expect a tuned Candidate 2 classifier in Phase 2 to reach balanced accuracy in the high 60s and PR-AUC near 0.40–0.45, and Candidate 1's regression to reach MAE of roughly 3.5–4 days. That is where the ceiling honestly sits: the largest driver of both outcomes – month-to-month carrier conditions – is not observable at prediction time. A late-delivery classifier reporting 95% accuracy would be either the majority-class rule or the leaked variant above.

3 Dashboard and Repository

Dashboard (Streamlit): `https://it5006-group-project-10.streamlit.app`
Repository (GitHub): `https://github.com/gengyuzhu/IT5006-Group-Project-10`

Both read through the same `src/data_load.py` entry point as this report, and `src/report_stats.py` regenerates every number quoted above.

References

[1] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. doi: 10.1016/j.ins.2011.12.028.

[2] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

[3] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.

[4] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

[5] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.

[6] João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4), 2014. doi: 10.1145/2523813.

[7] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022) Datasets and Benchmarks Track*, 2022.

[8] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001.

[9] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 2012. doi: 10.1145/2382577.2382579.

[10] Daniele Micci-Barreca. A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. *ACM SIGKDD Explorations Newsletter*, 3(1):27–32, 2001. doi: 10.1145/507533.507538.

[11] Olist and Sionek, André. Brazilian E-Commerce public dataset by Olist. Kaggle, 2018. Distributed for IT5006 via Canvas; nine relational CSV tables covering 99,441 orders placed between 2016 and 2018.

[12] A. Parasuraman, Valarie A. Zeithaml, and Arvind Malhotra. E-S-QUAL: A multiple-item scale for assessing electronic service quality. *Journal of Service Research*, 7(3):213–233, 2005. doi: 10.1177/1094670504271156.

[13] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[14] Shashank Rao, Thomas J. Goldsby, Stanley E. Griffis, and Deepak Iyengar. Electronic logistics service quality (e-LSQ): Its impact on the customer's purchase satisfaction and

retention. *Journal of Business Logistics*, 32(2):167–179, 2011. doi: 10.1111/j.2158-1592.2011.01014.x.

[15] Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3): e0118432, 2015. doi: 10.1371/journal.pone.0118432.

A Data Model, Audit and Dictionary

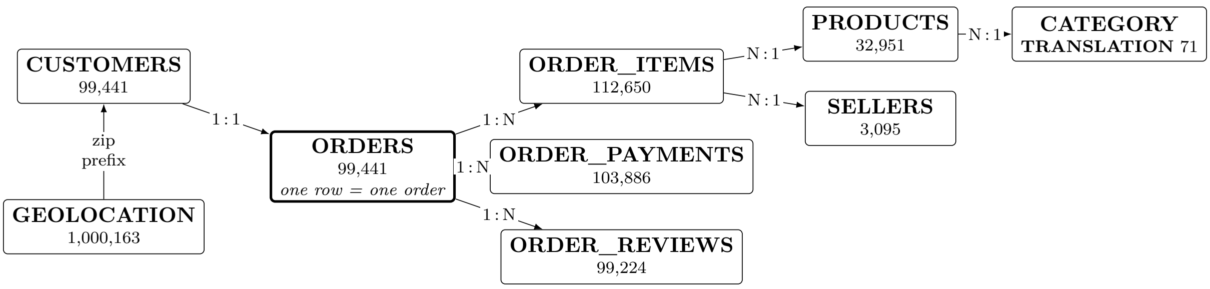

Figure 1: The nine source tables. ORDERS is the analysis grain. The three one-to-many children (bold arrows to items, payments and reviews) are each aggregated to one row per order *before* joining; merging them directly inflates the table by 19% and biases every average.

Table 1: Data quality audit and cleaning decisions, implemented as CLEANING_DECISIONS in the data-loading module.

Issue	Evidence	Decision
One-to-many joins	112,650 items and 103,886 payments against 99,441 orders; a naive merge yields 118,437 rows	Aggregate each child table to one row per order, then join; assert the grain
Customer key	99,441 customer_id values but 96,096 customer_unique_id; 2,997 people ordered more than once	Carry customer_unique_id and group on it for repeat behaviour and for splitting
Undelivered orders	2.98% of orders have no delivery timestamp; status is *delivered* for 97.0%	Delivery targets computed only for delivered orders; the rest are NaN, never imputed
Review keys	review_id is not unique (789 ids reused); 547 orders carry 2–3 reviews and 768 carry none	Key on order_id, keep the earliest review; no review means NaN
Sparse review text	review_comment_title present for 11.7% of reviews, review_comment_message for 41.3%	Use review_score only; coverage does not support NLP features
Geolocation grain	1,000,163 rows for 19,010 zip prefixes, including points outside Brazil	Collapse to a median coordinate per prefix; use it for great-circle distance
Missing product attributes	610 products (1.9%) have no category	Label unknown rather than dropping orders; impute numerics in-pipeline
Coverage ramp	326 orders before 2017; final weeks truncated mid-cycle	Restrict to 2017-01-01 .. 2018-08-31 (99,091 orders)
Portuguese categories	71 category names	Translate to English for all output; keep the original key

Table 2: Candidate problems against the brief's scoping checklist.

Criterion	C1 delivery (checkout)	C2 low review (delivery)	C3 repeat (checkout)
Derivable target	Yes, from order timestamps	Yes, `review_score` ≤ 2	Yes, next order by person
No leakage	Yes, checkout regime	Yes, at-delivery regime	Yes, checkout regime
Sufficient signal	Regression yes (-41% MAE); classification weak ($1.6\times$)	Yes, PR-AUC 0.40 vs 0.11 ($3.6\times$)	No, $2.0\times$ at a 1.7% base rate
Manageable imbalance	8.1% late; weights + PR-AUC	14.6% low; weights + PR-AUC	2.6%, too extreme
Clear stakeholder	Operations	Customer experience	CRM
Verdict	**Primary**	**Secondary**	Rejected

B Figures

Table 3: Data dictionary for the nine source tables, in the column-level format used by the NIBRS reference dictionary supplied with the brief. *Nullable* is measured, not declared: Y means the column actually contains missing values in this extract, with the observed rate given.

Table	Column	Type	Null	Comment
`orders`	`order_id`	text	N	Primary key. The analysis grain
	`customer_id`	text	N	**Per-order** key, not a person
	`order_status`	text	N	8 values; *delivered* for 97.0%
	`order_purchase_timestamp`	datetime	N	The checkout prediction moment
	`order_approved_at`	datetime	Y 0.16%	Post-checkout; excluded as a feature
	`order_delivered_carrier_date`	datetime	Y 1.79%	Post-outcome
	`order_delivered_customer_date`	datetime	Y 2.98%	Post-outcome; defines both delivery targets
	`order_estimated_delivery_date`	date	N	Promised at checkout; **admissible**. Date only, 459 distinct values
`order_items`	`order_id`	text	N	**1:N** — 112,650 rows for 98,666 orders
	`order_item_id`	int	N	Sequence within the order, 1–21
	`product_id`, `seller_id`	text	N	Foreign keys
	`shipping_limit_date`	datetime	N	Post-checkout; excluded
	`price`, `freight_value`	float	N	Per item; summed to order level
`order_payments`	`order_id`	text	N	**1:N**. One order has no payment row
	`payment_sequential`	int	N	1–29
	`payment_type`	text	N	5 values; credit card 75.4%
	`payment_installments`	int	N	0–24; > 1 for 51.4% of orders
	`payment_value`	float	N	Summed to order level
`order_reviews`	`review_id`	text	N	**Not unique** — 789 ids reused across orders
	`order_id`	text	N	**1:N**; 547 orders carry 2–3 reviews
	`review_score`	int	N	1–5; target of Candidate 2
	`review_comment_title`	text	Y 88.3%	Too sparse to use
	`review_comment_message`	text	Y 58.7%	Too sparse to use; no NLP features
	`review_creation_date`	date	N	Post-outcome for the review target
	`review_answer_timestamp`	datetime	N	Post-outcome
`customers`	`customer_id`	text	N	Joins 1:1 to orders
	`customer_unique_id`	text	N	**The actual person**; 96,096 of them. Grouping key for splits
	`customer_zip_code_prefix`	int	N	Joins to geolocation
	`customer_city`, `customer_state`	text	N	4,119 cities, 27 states
`sellers`	`seller_id`	text	N	3,095 sellers
	`seller_zip_code_prefix`	int	N	Joins to geolocation
	`seller_city`, `seller_state`	text	N	59.7% of sellers are in SP
`products`	`product_id`	text	N	32,951 products
	`product_category_name`	text	Y 1.85%	Portuguese; 73 values. Nulls labelled *unknown*
	`product_weight_g`	float	Y 0.01%	Summed to order level
	`product_length/height/width_cm`	float	Y 0.01%	Combined into item volume
	`product_photos_qty`	float	Y 1.85%	Listing-quality proxy
`geolocation`	`geolocation_zip_code_prefix`	int	N	19,015 prefixes over 1,000,163 rows
	`geolocation_lat`, `_lng`	float	N	Collapsed to a median per prefix; out-of-Brazil points dropped
`category_translation`	`product_category_name`	text	N	Join key
	`product_category_name_english`	text	N	71 translations
Derived in `src/` — not present in any source file:				
derived	`delivery_days`	float	Y	Purchase to arrival; target of C1A
derived	`is_late`	bool	Y	Arrival after the promised date; target of C1B
derived	`distance_km`	float	Y 1.2%	Great-circle customer–seller distance
derived	`estimated_days`	float	N	Promised window in days
derived	`lead_category`	text	N	Category of the most expensive item
derived	`freight_ratio`	float	Y	Freight as a share of item value

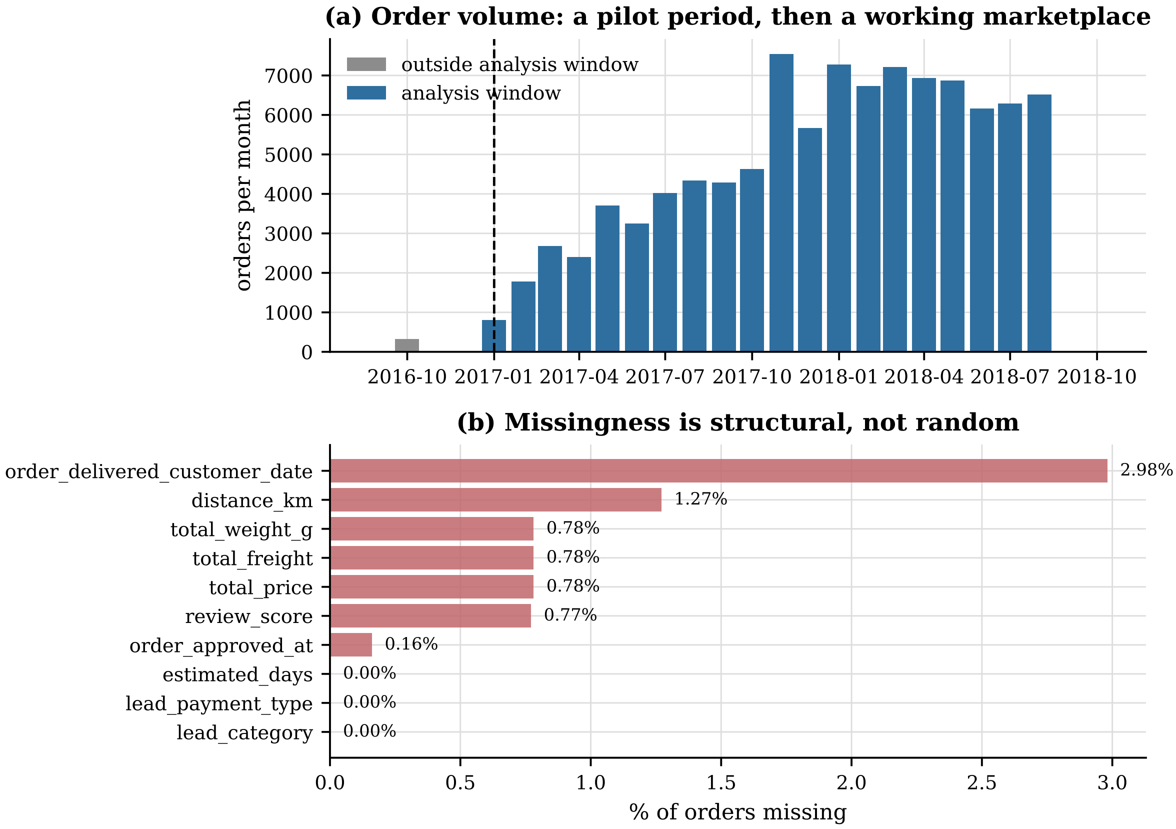

Figure 2: (a) Monthly order volume. The pilot period before 2017 contributes 326 orders in total and is excluded from the analysis window. (b) Missingness by column: structural rather than random, dominated by orders that never reached the customer.

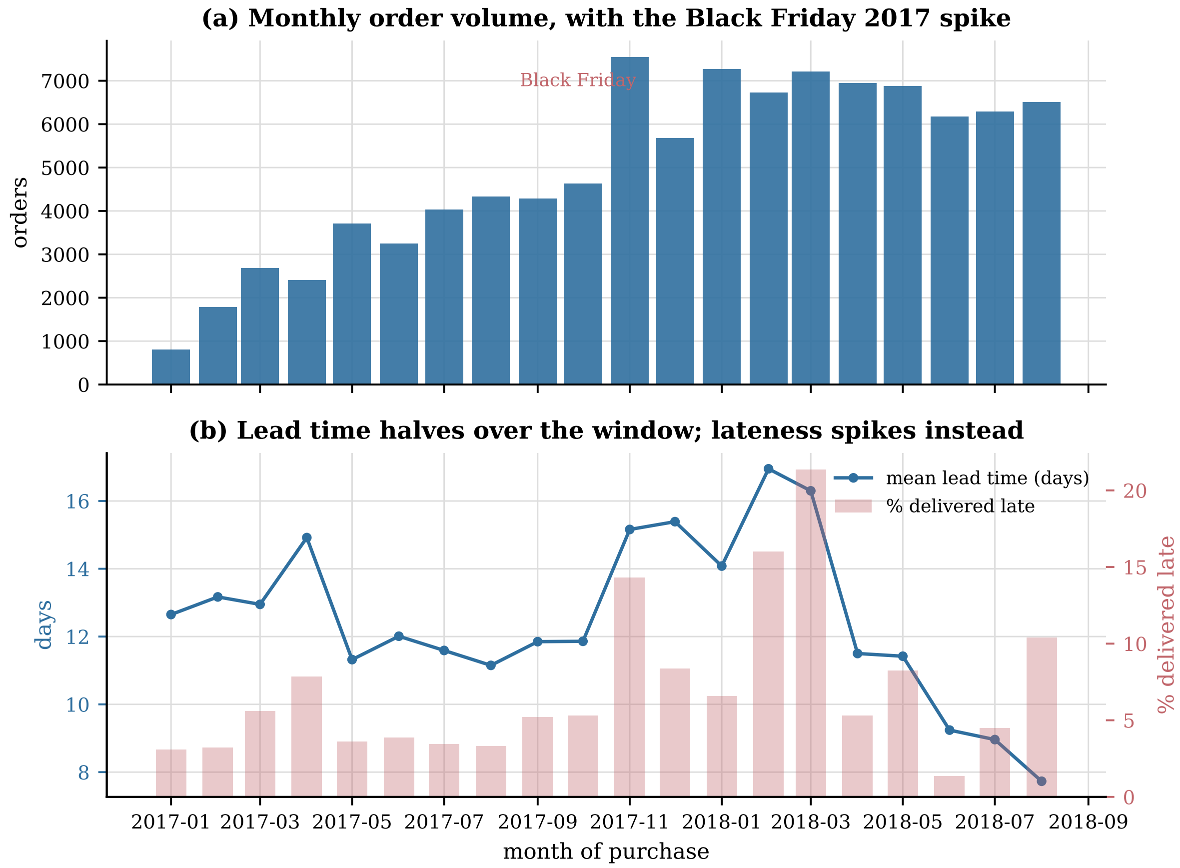

Figure 3: (a) Order volume with the Black Friday 2017 spike. (b) The central temporal finding: mean delivery lead time halves across the window (line, left axis) while the late rate is volatile rather than trending (bars, right axis), spiking at Black Friday and during the May 2018 truckers' strike.

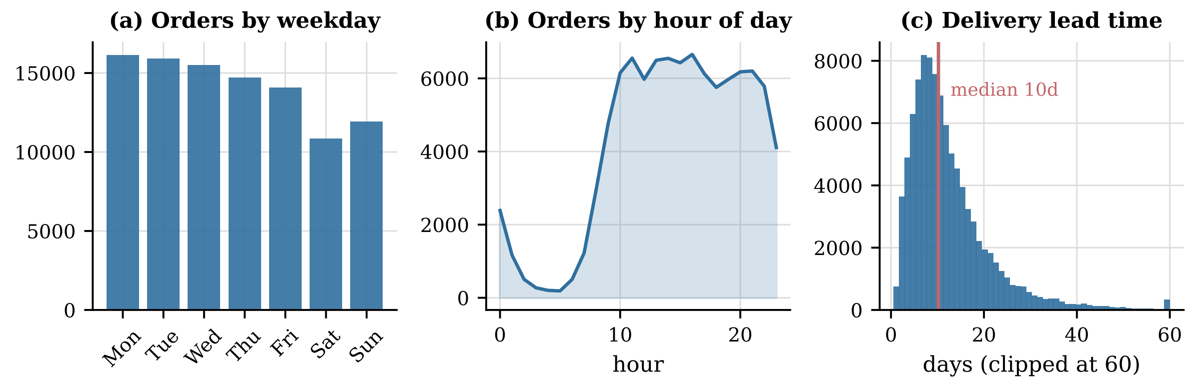

Figure 4: Purchase rhythm and the lead-time distribution. Ordering is weekday-heavy and peaks at 16:00; lead time is right-skewed with a median of 10.2 days and a long tail.

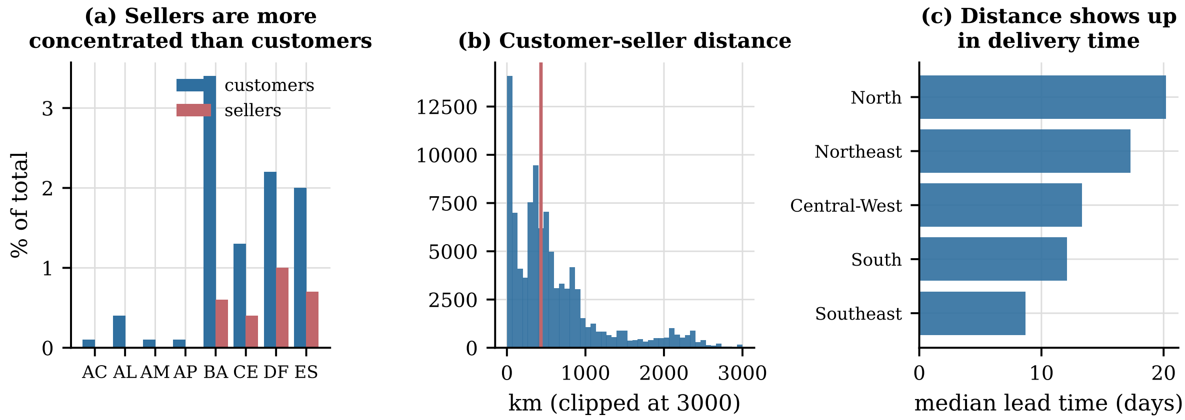

Figure 5: (a) Sellers are far more concentrated in São Paulo than customers are, so most orders must cross the country. (b) Customer–seller distance. (c) That distance shows up directly in regional delivery times.

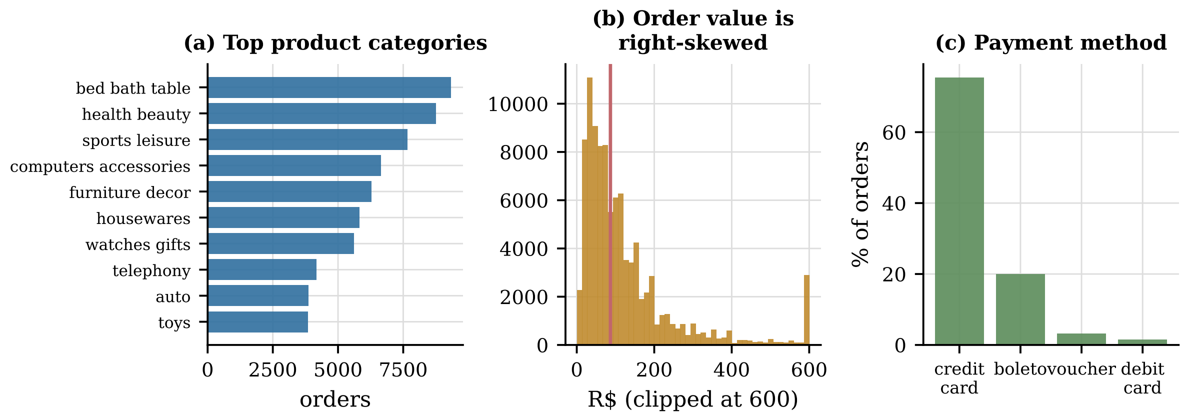

Figure 6: (a) Demand is long-tailed across 74 categories. (b) Order value is strongly right-skewed. (c) Credit card dominates, usually in instalments.

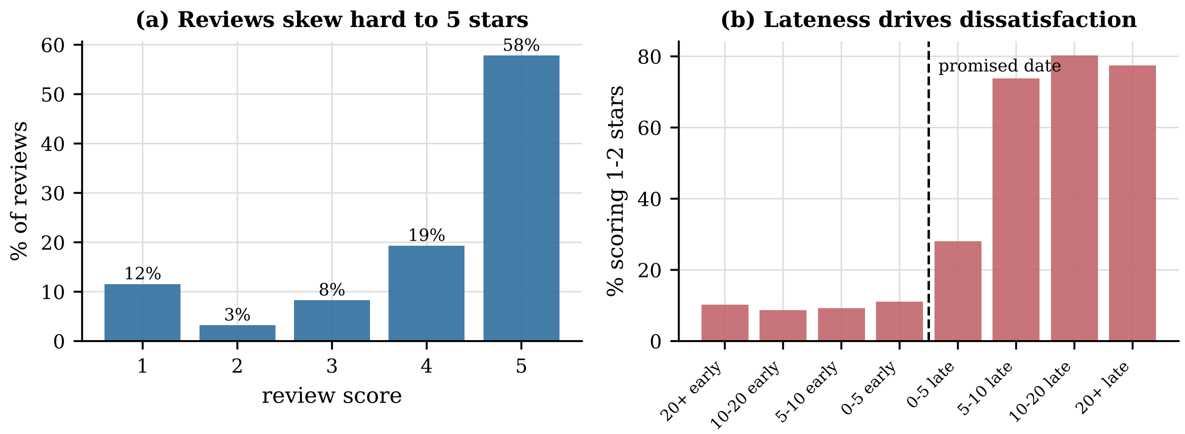

Figure 7: (a) Reviews are severely imbalanced towards five stars. (b) The strongest relationship in the dataset: the 1–2 star rate rises monotonically with lateness, from 9.2% for on-time orders to 54.0% for late ones.

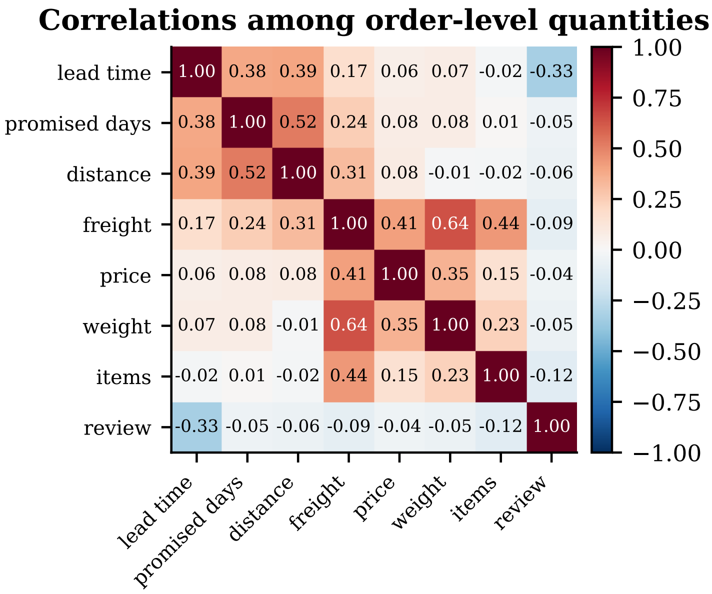

Figure 8: Correlations among order-level quantities. Distance and the platform's own promise are the strongest checkout-time correlates of lead time.

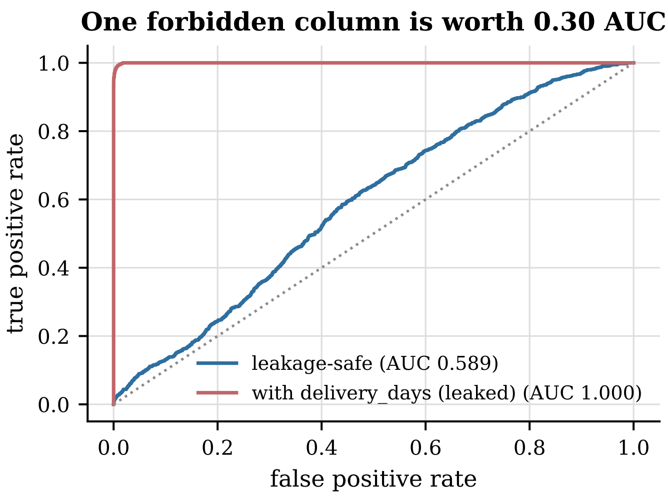

Figure 9: What one forbidden column is worth. Adding the realised delivery time to the late-delivery model takes ROC-AUC from 0.589 to 1.000 — the model is reading the answer, not predicting it.

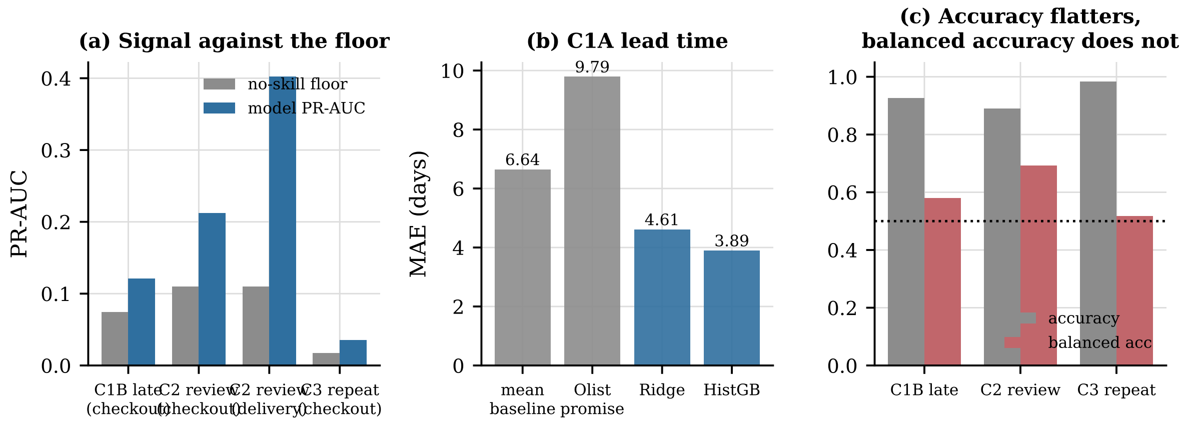

Figure 10: Candidate comparison. (a) PR-AUC against the no-skill floor for each candidate and prediction moment. (b) Lead-time MAE: the platform's own promise is worse than a constant. (c) Raw accuracy flatters every imbalanced target; balanced accuracy does not.